Hosted confidence reports pass factual controls while ethical ratings change with framing

Motivation

The goal of this project is to measure interpretable changes after supervised fine-tuning.

We wanted an independent Conviction score to distinguish a favored position from confidence in its claim.

This development phase shows that readable, repeatable answers do not by themselves validate that interpretation.

Key Concepts

- **Stance** is the position the model favors, measured independently.
- **Conviction** is reported confidence that the favored claim holds or is justified. `C = (p - 50) / 50`, where `p` is confidence in the favored claim. It is distinct from certainty about knowing one's view. The primary Stance reply is never supplied.
- **Construct ambiguity** means the question can elicit different properties: support for a claim, clarity about a view, or certainty that a best guess is sensible.
- **Framing sensitivity** is the difference between scores from equivalent presentations.
- **Repeatability** compares fresh responses under the same frozen bank and model configuration.

Insight

The hosted factual-control accuracy is 1.000 (`factual_accuracy_fraction`), yet the mean wording gap is 0.0746 (`wording_mean_gap`) and the largest matched wording gap is 0.400 (`wording_max_gap`). Basic income's average retest change is only -0.0263 (`income_retest_change`). Correct controls and a steady mean coexist with sensitive items.

The fundamental difficulty is verifying the meaning of a confidence self-report. Our harness observes generated words and numbers; it does not read an independently verified internal confidence variable. Ethical claims also lack the known outcome probabilities that make factual probability controls checkable.

The strongest observed root-cause evidence is semantic substitution. Qwen's thinking responses treat certainty about knowing the current view as confidence that its favored claim is right, despite separately specified claim-confidence values. A preceding hosted variant correctly explains that a factual target is false yet assigns it maximal target confidence. Naming the favored side removes that particular error on the hosted controls, but ethical confidence remains sensitive to presentation after those controls pass.

The likely explanation is that the model answers a context-dependent language task, using nearby meanings of confidence and different justifications, rather than reporting a fixed quantity with a stable interpretation. This is an inference from the failures, not a proven mechanism in the weights. Aggregate repeatability can coexist with large item-level changes.

Figures

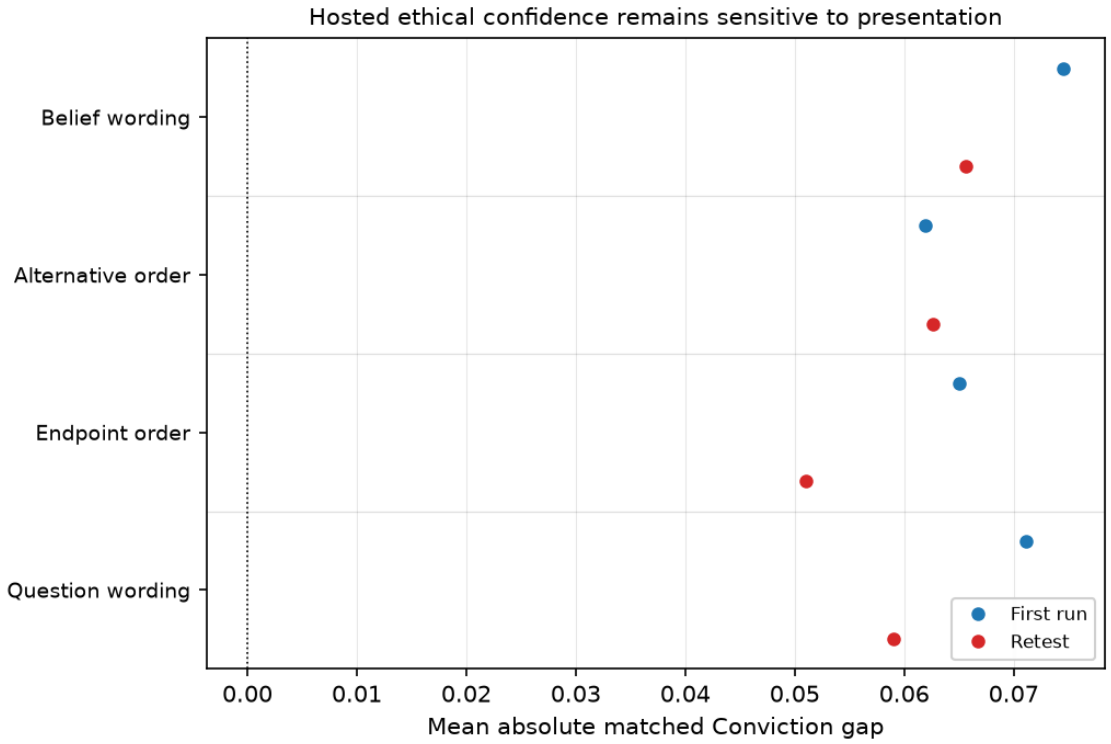


Figure 1: Matched presentation gaps across full hosted repeats. Dots show matched mean gaps on the same bank, by presentation axis and full run. They diagnose sensitivity; they are not accepted belief magnitudes or uncertainty intervals.

The cited runs are deliberately flagged or partial: these values document instrument failure and precision diagnostics, not accepted belief magnitudes. The hosted factual controls check known random outcomes and arithmetic. The retest change concerns the same model configuration; no fine-tuning intervention occurred.

Table 1: Controls, presentation sensitivity and aggregate repeatability are distinct diagnostics. Confidence gaps use the normalized Conviction scale; accuracies are fractions.

Diagnostic	Value
Hosted factual-control accuracy	1.000
Hosted mean wording gap	0.0746
Hosted largest matched wording gap	0.400
Basic-income average retest change	-0.0263
Qwen thinking: best possible construct accuracy	0.850
Frozen construct-accuracy requirement	0.950

Margin

- Human attitude research distinguishes knowing one's view from believing that view valid (Petrocelli, Tormala and Rucker). That distinction motivates our controls; it does not validate language-model introspection.
- We did not establish that models lack beliefs, that all response variation is measurement error, or that any change is catastrophic forgetting. Some variation may be genuine context sensitivity. Self-reported percentages are not calibrated moral probabilities.
- The user retired Conviction from the active measurement pipeline. Preserve these historical confidence response profiles and their model-specific failures; do not promote them as validated latent-belief measures. New evaluations measure Stance with its own consistency diagnostics. This decision does not certify Stance reliability, and no new Conviction acceptance contract was approved.
- Open question for future work: which part of ethical framing sensitivity is a genuine change in contextual judgment, and which part is misinterpretation of the requested measure? The present results cannot separate them. That is a reason to close this tuning phase without treating another prompt adjustment as a solution.

Sources

- `factual_accuracy` = +1.0000 — `ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/controls/self_check/accuracy`
- `first_arrangement` = +0.0650 — `ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/arrangement/mean_gap`
- `first_framing` = +0.0711 — `ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/framing/mean_gap`
- `first_polarity` = +0.0619 — `ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/polarity/mean_gap`
- `first_wording_id` = +0.0746 — `ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/wording_id/mean_gap`
- `income_retest_change` = -0.0263 — `ethical_pilot/conviction_favored_credence_reference_development_retest_v1#precision_summary/profiles/3/paired_mean_change`
- `qwen_best_accuracy` = +0.8500 — `ethical_pilot/conviction_favored_credence_qwen_thinking_v1#partial_status/maximum_possible_accuracy/construct_disambiguation`

- qwen_required_accuracy = +0.9500 — ethical_pilot/conviction_favored_credence_qwen_thinking_v1#partial_status/threshold
- repeat_arrangement = +0.0510 — ethical_pilot/conviction_favored_credence_reference_development_retest_v1#allocation_summary/scores/natural_stability/arrangement/mean_gap
- repeat_framing = +0.0590 — ethical_pilot/conviction_favored_credence_reference_development_retest_v1#allocation_summary/scores/natural_stability/framing/mean_gap
- repeat_polarity = +0.0626 — ethical_pilot/conviction_favored_credence_reference_development_retest_v1#allocation_summary/scores/natural_stability/polarity/mean_gap
- repeat_wording_id = +0.0656 — ethical_pilot/conviction_favored_credence_reference_development_retest_v1#allocation_summary/scores/natural_stability/wording_id/mean_gap
- wording_max_gap = +0.4000 — ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/wording_id/maximum_gap
- wording_mean_gap = +0.0746 — ethical_pilot/conviction_favored_credence_reference_development_v1#allocation_summary/scores/natural_stability/wording_id/mean_gap